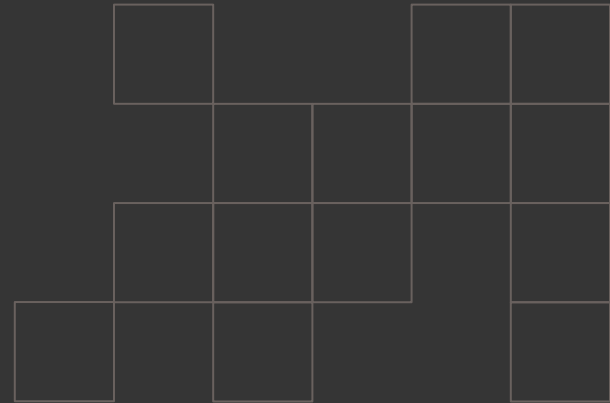


DATS 598
Fall 2025

Statistical Truth Detection System

Author: Eric Crisp



Determine the value added by adding truth detection in LLM output

Prompting an LLM should come with a degree of confidence in the result with verifiable evidence.

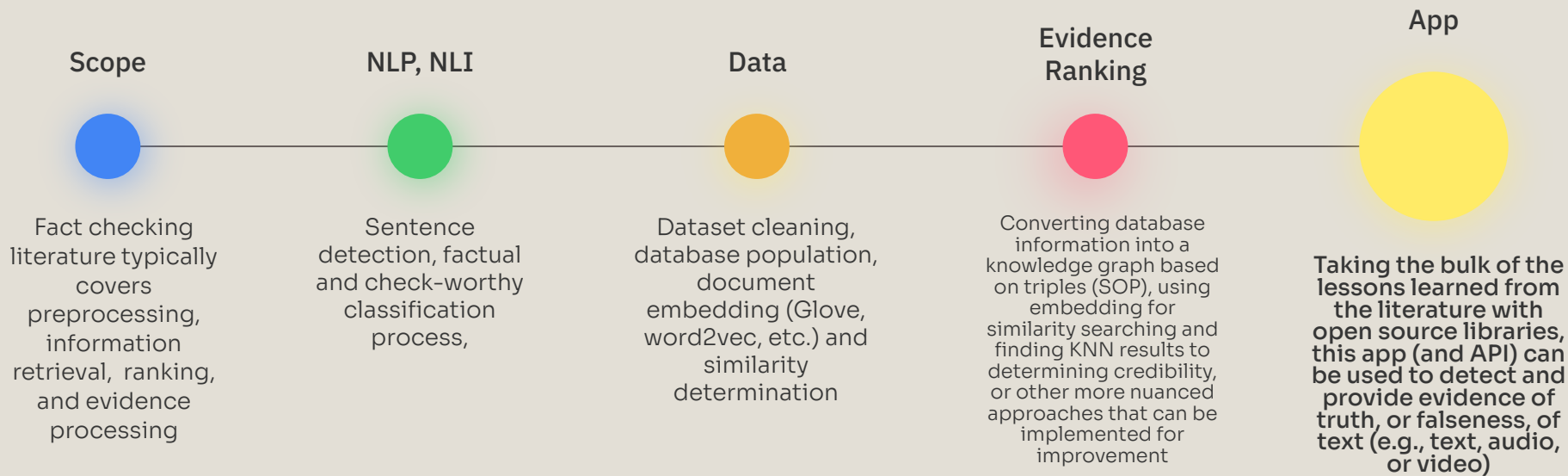
Claude, GPT, Gemini have fact-checking incorporated. If you turn it off, what happens?

Can simpler LLMs show better results with other evidence-based systems incorporated?

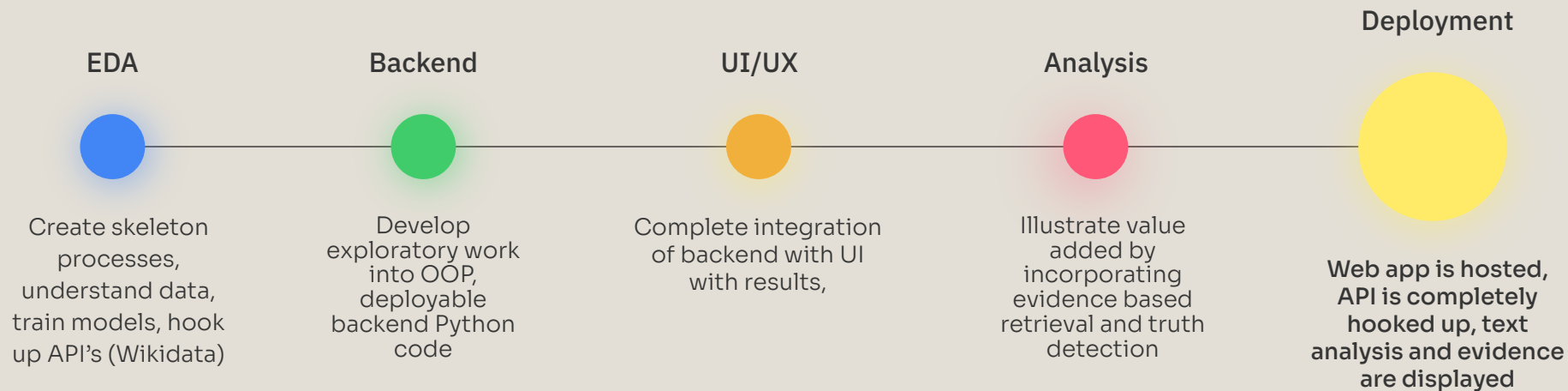
Through conversion and preprocessing of text input, evidence retrieval, similarity and overlap analysis, as well as evidence credibility, the truthiness of LLM output can be evaluated

Open source embeddings, LMs, machine learning algorithms, and deep networks are all utilized in this process

Literature Review



(Remaining) Roadmap & Milestones



Resources

Contact list



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Links to key files and documents

Repository

<https://github.com/ericmcrisp/dats598>

Any questions?
Ask away!



EXTRA
SLIDES

Predicting Upcoming Challenges

Potential challenge

Projected Response

1

NLP fact detection and discerning from opinion consistently

Leverage open source libraries for forming industry standard (e.g., from the literature) cleaning/processing prior to prompt generation and search

2

RAG, evidence retrieval, crawling for real-time, accurate responses

Create toy model for crawling and understand scope prior to incorporating functionality (a nice to have)

3

Determining probability of truth from inconsistently trustworthy sources

Apply a likelihood based on sources, overlap, similarity of document to text to determine how likely the content from a source is based on priors